



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

15-06-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	B	A	A	A	C	C	C	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	D	A	D	A	C	D	D	C
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	600	20	15	5	4	0	5	82	30

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	B	B	C	B	C	D	B	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	B	B	C	B	A	D	D	D
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	20	3	16	106	2	9	9	50	33	10

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	D	B	C	D	B	B	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	C	B	C	C	D	D	B	C
SECTION-III	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3	1	1	4	766	36	406	2	16	4

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (D)

$$(n-1)a = n(a')$$

$$a' = \frac{(n-1)a}{n}$$

$$\therefore \text{L.C.} = 1 \text{ MSD} - 1 \text{ VSD}$$

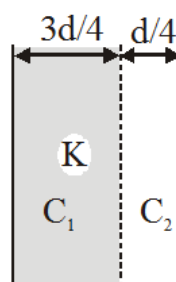
$$= (a - a') \text{ cm}$$

$$= a - \frac{(n-1)a}{n}$$

$$= \frac{na - na + a}{n} = \frac{a}{n} \text{ cm}$$

$$= \left(\frac{10a}{n} \right) \text{ mm}$$

2. Ans (C)



$$C_0 = \frac{\epsilon_0 A}{d}$$

$C' = C_1$ and C_2 in series.

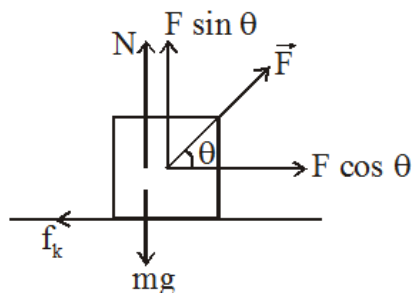
$$\text{i.e. } \frac{1}{C'} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C'} = \frac{(3d/4)}{\epsilon_0 K A} + \frac{d/4}{\epsilon_0 A}$$

$$\frac{1}{C'} = \frac{d}{4\epsilon_0 A} \left(\frac{3+K}{K} \right)$$

$$C' = \frac{4KC_0}{(3+K)}$$

3. Ans (B)



$$N = mg - F \sin \theta$$

$$F \cos \theta - \mu_k N = ma$$

$$F \cos \theta - \mu_k (mg - F \sin \theta) = ma$$

$$a = \frac{F}{m} \cos \theta - \mu_k \left(g - \frac{F}{m} \sin \theta \right)$$

4. Ans (A)

$$P_1 = \rho g d + P_0 = 3 \times 10^5 \text{ Pa}$$

$$\therefore \rho g d = 2 \times 10^5 \text{ Pa}$$

$$P_2 = 2\rho g d + P_0$$

$$= 4 \times 10^5 + 10^5 = 5 \times 10^5 \text{ Pa}$$

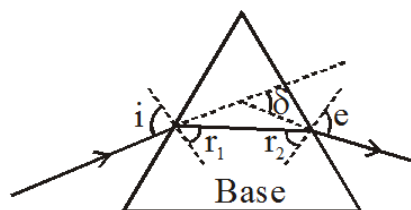
$$\% \text{ increase} = \frac{P_2 - P_1}{P_1} \times 100$$

$$= \frac{5 \times 10^5 - 3 \times 10^5}{3 \times 10^5} \times 100 = \frac{200}{3} \%$$

5. Ans (A)

Deviation is minimum in a prism when :

$i = e$, $r_1 = r_2$ and ray (2) is parallel to base of prism.



6. Ans (A)

$$f = 5 \times 10^8 \text{ Hz}$$

EM wave is travelling towards $+\hat{j}$

$$\vec{B} = 8.0 \times 10^{-8} \hat{z} \text{ T}$$

$$\vec{E} = \vec{B} \times \vec{C} = (8 \times 10^{-8} \hat{z}) \times (3 \times 10^8 \hat{y})$$

$$= -24 \hat{x} \text{ V/m}$$

7. Ans (C)

By angular momentum conservation :

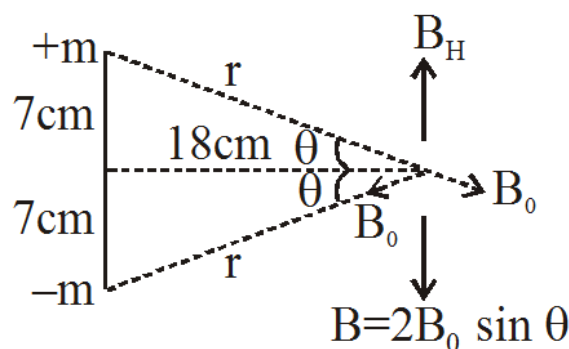
$$mv_1 r_1 = mv_2 r_2$$

$$v_1 = \frac{48 \times 10^{14}}{1.6 \times 10^{12}} = 3000 \text{ m/sec}$$

$$= 3 \times 10^3 \text{ m/sec.}$$

8.

Ans (C)



$$\text{i.e. } \frac{2\mu_0}{4\pi} \frac{m}{r^2} \times \frac{7}{r} = 0.4 \times 10^{-4}$$

$$\Rightarrow 2 \times 10^{-7} \times \frac{m \times 7}{(7^2 + 18^2)^{3/2}} \times 10^4$$

$$= 0.4 \times 10^{-4}$$

$$m = \frac{4 \times 10^{-2} \times (373)^{3/2}}{14}$$

$$M = m \times 14 \text{ cm} = m \times \frac{14}{100}$$

$$= \frac{0.04 \times (373)^{3/2}}{14} \times \frac{14}{100}$$

$$= 4 \times 10^{-4} \times 7203.82 = 2.88 \text{ J/T}$$

9.

Ans (C)

$$PV = (n_1 + n_2 + n_3)RT$$

$$P \times V = \left[\frac{16}{32} + \frac{28}{28} + \frac{44}{44} \right] RT$$

$$PV = \left[\frac{1}{2} + 1 + 1 \right] RT$$

$$P = \frac{5}{2} \frac{RT}{V}$$

10.

Ans (A)

Heat and work are treated as path functions in thermodynamics.

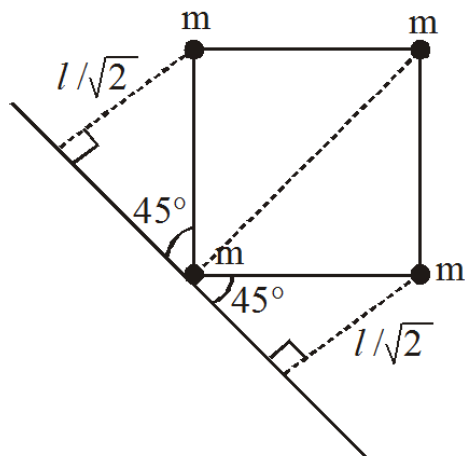
$$\Delta Q = \Delta U + \Delta W$$

Since work done by gas depends on type of process i.e. path and ΔU depends just on initial and final states, so ΔQ i.e. heat, also has to depend on process is path.

11. **Ans (C)**

Moment of inertia of point mass

$$= \text{mass} \times (\text{Perpendicular distance from axis})^2$$



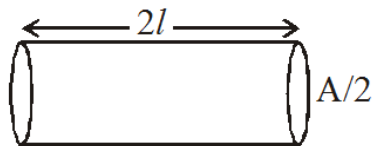
Moment of Inertia

$$= m(0)^2 + m(\ell\sqrt{2})^2 + m\left(\frac{\ell}{\sqrt{2}}\right)^2 + m\left(\frac{\ell}{\sqrt{2}}\right)^2$$

$$= 3m\ell^2$$

12. **Ans (A)**

As per the question



$$\text{Resistance} = \frac{\rho(2\ell)}{(A/2)} = \frac{4\rho\ell}{A}$$

$$\Rightarrow \text{Current} = \frac{V}{R} = \frac{VA}{4\rho\ell}$$

13. **Ans (D)**

When lift is stationary

$$T = 2\pi\sqrt{\frac{L}{g}}$$

When lift is moving upwards

$\Rightarrow$ Pseudo force acts downwards

$$\Rightarrow g_{\text{eff}} = g + \frac{g}{2} = \frac{3g}{2}$$

$\Rightarrow$ New time period

$$T' = 2\pi\sqrt{\frac{L}{g_{\text{eff}}}} = 2\pi\sqrt{\frac{2L}{3g}}$$

$$T' = \sqrt{\frac{2}{3}} T$$

14. **Ans (A)**

For $0 \leq x \leq 200$

$$v = mx + C$$

$$v = \frac{1}{5}x + 10$$

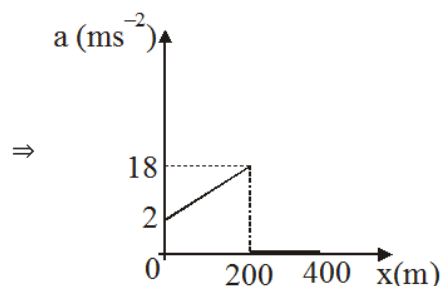
$$a = \frac{v dv}{dx} = \left(\frac{x}{5} + 10\right) \left(\frac{1}{5}\right)$$

$$a = \frac{x}{25} + 2 \Rightarrow \text{Straight line till } x = 200$$

for $x > 200$

$$v = \text{constant}$$

$$\Rightarrow a = 0$$



Hence most appropriate option will be (1), otherwise it would be BONUS.

15. **Ans (D)**

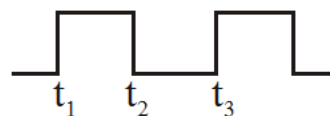
$$\text{Length of Antena} = 25\text{m} = \frac{\lambda}{4}$$

$$\Rightarrow \lambda = 100\text{ m}$$

16. **Ans (A)**

In EMW, Average energy density due to electric (U_e) and magnetic (U_m) fields is same.

17. **Ans (C)**



For $t_1 - t_2$ Charging graph

$t_2 - t_3$ Discharging graph

18. **Ans (D)**

Stopping potential changes linearly with frequency of incident radiation.

19. **Ans (D)**

$$N = m\omega^2 R$$

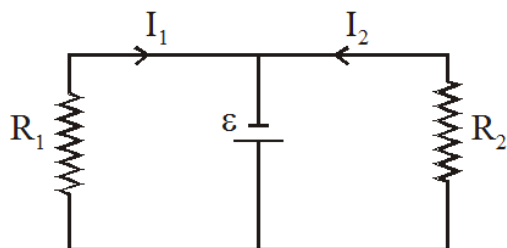
$$N = m \left[\frac{4\pi^2}{T^2} \right] R \quad \dots(1)$$

Given $m = 0.2\text{ kg}$, $T = 40\text{ S}$, $R = 0.2\text{ m}$

Put values in equation (1)

$$N = 9.859 \times 10^{-4}\text{ N}$$

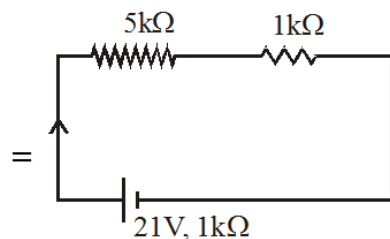
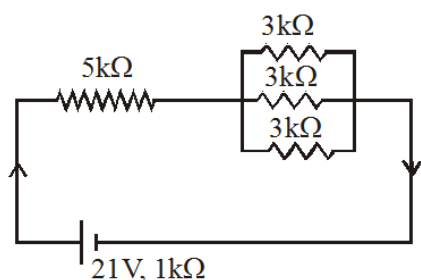
20. Ans (C)



PART-1 : PHYSICS

SECTION-II

1. Ans (3)



$$I = \frac{21}{5 + 1 + 1} = 3 \text{ mA}$$

2. Ans (600)

$$\beta = \frac{\lambda D}{d}$$

$$\lambda = \frac{\beta d}{D}$$

$$\lambda = \frac{6 \times 10^{-3} \times 10^{-3}}{10}$$

$$\lambda = 6 \times 10^{-7} \text{ m} = 600 \times 10^{-9} \text{ m}$$

$$\lambda = 600 \text{ nm}$$

3. Ans (20)

$$\alpha = \frac{\tau}{I} = \frac{F \cdot R}{mR^2/2} = \frac{2F}{mR}$$

$$\alpha = \frac{2 \times 200}{20 \times (0.2)} = 10 \text{ rad/s}^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$$

$$(50)^2 = 0^2 + 2(10)\Delta\theta \Rightarrow \Delta\theta = \frac{2500}{20}$$

$$\Delta\theta = 125 \text{ rad}$$

$$\text{No. of revolution} = \frac{125}{2\pi} \approx 20 \text{ revolution}$$

4. Ans (15)

For 1st line

$$\frac{1}{\lambda_1} = R_Z^2 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)$$

$$\frac{1}{\lambda_1} = R_Z^2 \frac{5}{36} \dots\dots(i)$$

For 3rd line

$$\frac{1}{\lambda_3} = R_Z^2 \left(\frac{1}{2^2} - \frac{1}{5^2} \right)$$

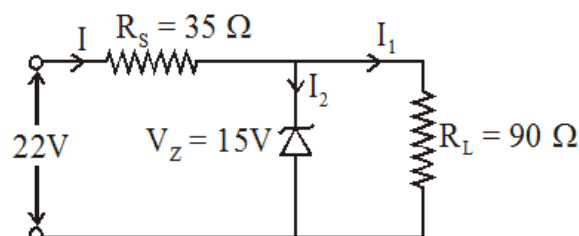
$$\frac{1}{\lambda_3} = R_Z^2 \frac{21}{100} \dots\dots(ii)$$

(ii) + (i)

$$\frac{\lambda_1}{\lambda_3} = \frac{21}{100} \times \frac{36}{5} = 1.512 = 15.12 \times 10^{-1}$$

$$x \approx 15$$

5. Ans (5)



Voltage across $R_S = 22 - 15 = 7\text{V}$

$$\text{Current through } R_S = I = \frac{7}{35} = \frac{1}{5} \text{ A}$$

$$\text{Current through } 90\Omega = I_2 = \frac{15}{90} = \frac{1}{6} \text{ A}$$

$$\text{Current through zener} = \frac{1}{5} - \frac{1}{6} = \frac{1}{30} \text{ A}$$

Power through zener diode

$$P = VI$$

$$P = 15 \times \frac{1}{30} = 0.5 \text{ watt}$$

$$P = 5 \times 10^{-1} \text{ watt}$$

6. Ans (4)

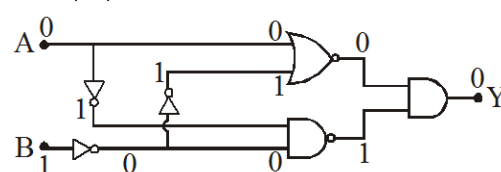
At resonance power (P)

$$P = \frac{(V_{\text{rms}})^2}{R}$$

$$P = \frac{(250/\sqrt{2})^2}{8} = 3906.25 \text{ W}$$

$$\approx 4 \text{ kW}$$

7. Ans (0)



8. Ans (5)

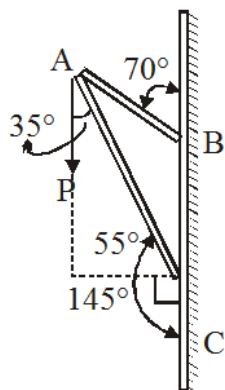
$$\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta I}{I} \times 100$$

$$\% \text{ error in } R = \frac{2}{50} \times 100 + \frac{0.2}{20} \times 100$$

$$\% \text{ error in } R = 4 + 1$$

$$\% \text{ error in } R = 5\%$$

9. Ans (82)



Component along AC

$$= 100 \cos 35^\circ$$

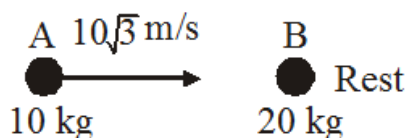
$$= 100 \times 0.819$$

$$= 81.9 \text{ N}$$

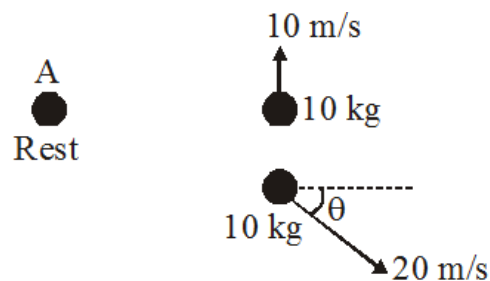
$$\approx 82 \text{ N}$$

10. Ans (30)

Before Collision



After Collision



From conservation of momentum along x axis;

$$\vec{P}_i = \vec{P}_f$$

$$10 \times 10\sqrt{3} = 200 \cos \theta$$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)

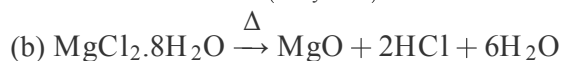
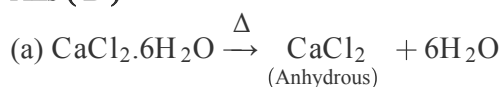
Size of ${}_{97}\text{Bk}^{3+}$ ion is less than that of ${}_{93}\text{Np}^{3+}$ due to actinoid contraction.

As we know that in a period from left to right ionic radius decreases and in actinide series it is due to actinoid contraction.

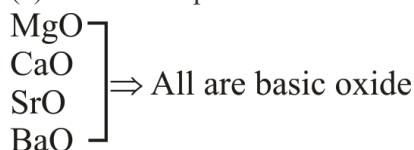
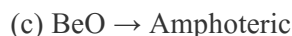
2. Ans (B)

Vitamin-A & Vitamin-D

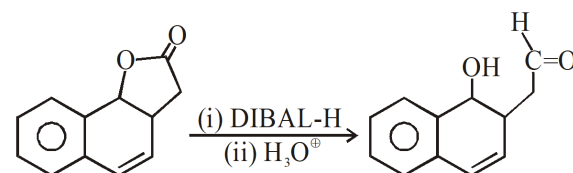
3. Ans (B)



The dehydration of hydrated chloride of calcium can be achieved. The corresponding hydrated chloride of magnesium on heating suffer hydrolysis.



4. Ans (B)



DIBAL can not reduce double bond

It can reduce cyclic ester.

5. Ans (C)

(a) Haber's process is used for NH_3 synthesis.

(b) Ostwald's process is used for HNO_3 synthesis.

(c) Contact process is used for H_2SO_4 synthesis.

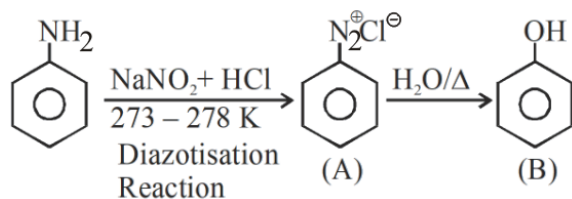
(d) In Hall-Heroult process, electrolytic reduction of impure alumina can be done. (Aluminium extraction)

6. Ans (B)

(A) Non-Aromatic (B) Aromatic

(C) Aromatic (D) Anti-Aromatic

7. **Ans (C)**



8. **Ans (D)**

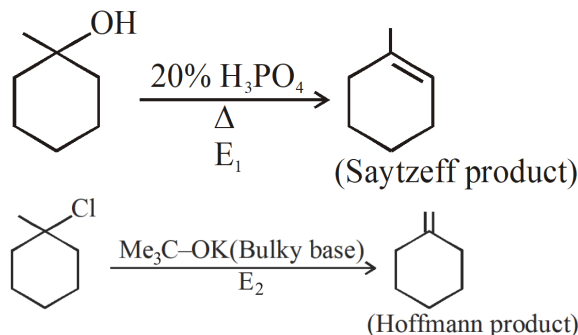
The E° value for Ce^{4+}/Ce^{3+} is +1.74 V because the most stable oxidation state of lanthanide series elements is +3.

It means Ce^{3+} is more stable than Ce^{4+} .

10. **Ans (D)**

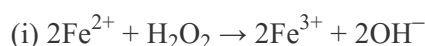
Partially deactivated palladised charcoal ($H_2/pd/CaCO_3$) is lindlar catalyst.

11. **Ans (C)**



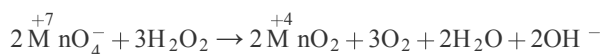
12. **Ans (B)**

(a) H_2O_2 can act as both oxidising and reducing agent in basic medium.



In this reaction, H_2O_2 acts as oxidising agent.

(ii)



In this reaction, H_2O_2 acts as reducing agent.

(b) The basic principle of hydrogen economy is the transportation and storage of energy in the form of liquids or gaseous dihydrogen.

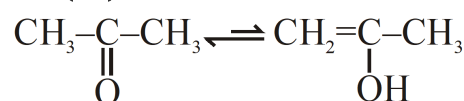
Advantage of hydrogen economy is that energy is transmitted in the form of dihydrogen and not as electric power

13. **Ans (B)**

In presence of ozone (O_3), oxidising smog gets increased during the day time because automobiles and factories produce main components of the photochemical smog (oxidising smog) results from the action of sunlight on unsaturated hydrocarbon and nitrogen oxide.

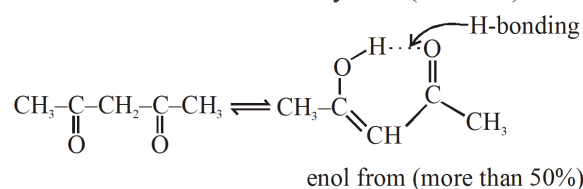
Ozone is strong oxidising agent and can react with the unburnt hydrocarbons in the polluted air to produce chemicals.

14. **Ans (B)**

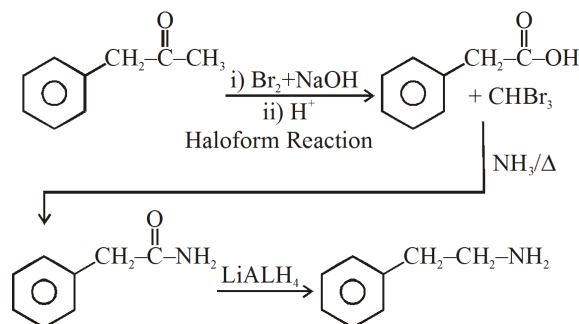


(Keto form) (enol form)

enol form of acetone is very less (< 0.1 %)



15. **Ans (C)**

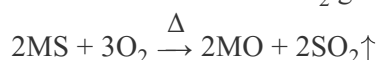


⇒ This reaction does not involve haffmann bromamide degradation.

⇒ Rest all options involve haffmann bromamide degradation during the reaction of $Br_2 + NaOH$ with amide.

16. **Ans (B)**

In roasting process, metal sulphide (MS) ore are converted into metal oxide and sulphur is removed in the form of SO_2 gas.



17. **Ans (A)**

(a) Hypophosphorus acid : H_3PO_2

$$(+1)3 + x + (-2)2 = 0$$

$$x = +1$$

(b) Orthophosphoric acid : H_3PO_4

$$(+1)3 + x + (-2)4 = 0$$

$$x = +5$$

(c) Hypophosphoric acid : $\text{H}_4\text{P}_2\text{O}_6$

$$(+1)4 + 2x + (-2)6 = 0$$

$$x = +4$$

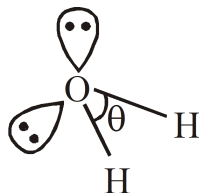
(d) Orthophosphorous acid : H_3PO_3

$$(+1)3 + x + (-2)3 = 0$$

$$x = +3$$

18. **Ans (D)**

H_2O



$$\theta = 104.5^\circ$$

the hybridisation of oxygen in water molecule is sp^3 .

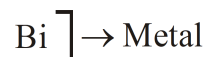
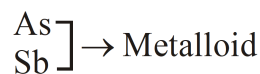
So electron geometry of water molecule is tetrahedral and the bond angle should be $109^\circ 28'$ but as we know that lone pair-lone pair repulsion of electrons is higher than the bond pair-bond pair repulsion because lone pair is occupied more space around central atom than that of bond pair.

19. **Ans (D)**

In chromatography technique, the purification of a compound is independent of the physical state of the pure compound.

20. **Ans (D)**

In group 15



Hydrides of group 15 elements are



In NH_3 , hydrogen atom gets partial positive charge due to less electronegativity.

But in BiH_3 , hydrogen atom gets partial negative charge because hydrogen is more electronegative than bismuth.

i.e. BiH_3 is a strong reducing agent than others because we know that H^- is a strong reducing agent.

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (20)**

$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

$$\text{Given } \Delta G^\circ = -9.478 \text{ KJ/mole}$$

$$T = 495\text{K} \quad R = 8.314 \text{ J mol}^{-1}$$

$$\text{So } -9.478 \times 10^3 = -495 \times 8.314 \times \ln K_{\text{eq}}$$

$$\ln K_{\text{eq}} = 2.303 = \ln 10$$

$$\text{So } K_{\text{eq}} = 10$$

$$\text{Now } \text{A(g)} \rightleftharpoons \text{B(g)}$$

$$t = 0 \quad 22 \quad 0$$

$$t = t \quad 22-x \quad x$$

$$K_{\text{eq}} = \frac{[\text{B}]}{[\text{C}]} = \frac{x}{22-x} = 10$$

$$\text{or } x = 20$$

$$\text{So millmoles of B} = 20.$$

2. Ans (3)

44 gm CO₂ have 12 gm carbon

$$\text{So, } 420 \text{ gm CO}_2 \Rightarrow \frac{12}{44} \times 420$$

$$\Rightarrow \frac{1260}{11} \text{ gm carbon}$$

$\Rightarrow 114.545 \text{ gram carbon}$

$$\text{So, \% of carbon} = \frac{114.545}{750} \times 100 \approx 15.3\%$$

18 gm H₂O $\Rightarrow$ 2 gm H₂

$$210 \text{ gm} \Rightarrow \frac{2}{18} \times 210$$

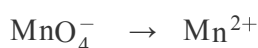
$= 23.33 \text{ gm H}_2$

$$\text{So, \% H}_2 \Rightarrow \frac{23.33}{750} \times 100 = 3.11\% \approx 3\%$$

3. Ans (16)

Writing the half reaction

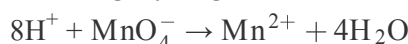
oxidation half reaction



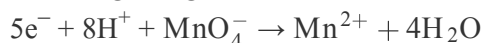
balancing oxygen



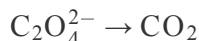
balancing Hydrogen



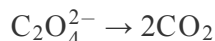
balancing charge



Reduction half



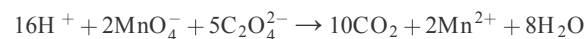
Balancing carbon



Balancing charge

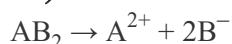


Net equation



So c = 16

4. Ans (106)



$$t=0 \quad a \quad 0 \quad 0$$

$$t=t \quad a-\alpha \quad \alpha \quad 2\alpha$$

$$n_T = a - \alpha + \alpha + 2\alpha = a(1 + 2\alpha)$$

$$\text{so } i = 1 + 2\alpha$$

$$\text{Now } \Delta T_b = i \times m \times K_b$$

$$\Delta T_b = (1 + 2\alpha) \times m \times K_b$$

$$\alpha = 0.1 \quad m = 10 \quad K_b = 0.5$$

$$\Delta T_b = 1.2 \times 10 \times 0.5 = 6$$

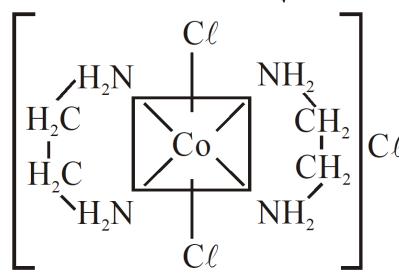
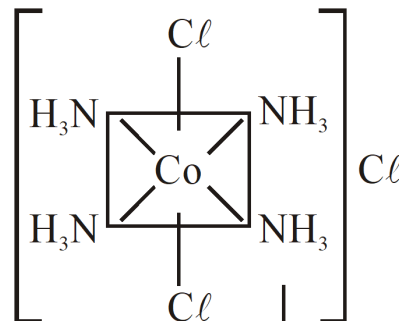
So boiling point = 106

5. Ans (2)

trans - CoCl₃.4NH₃

or

trans-[Co(NH₃)₄Cl₂]Cl



As we know that ethylene diamine is a bidentate ligand and ammonia is a monodentate ligand.

It means overall two ethylene diamine is required to replace the all neutral ligands (four ammonia) from the coordination sphere of this complex.

6. Ans (9)

6.5 molal KOH = 1000gm solvent has 6.5 moles KOH

$$\text{so wt of solute} = 6.5 \times 56 = 364 \text{ gm}$$

$$\text{wt of solution} = 1000 + 364 = 1364$$

$$\text{Volume of solution} = \frac{1364}{1.89} \text{ ml}$$

$$\text{Molarity} = \frac{\text{mole of solute}}{V_{\text{solution in Litre}}}$$

$$= \frac{6.5 \times 1.89 \times 1000}{1364} = 9.00$$

7. **Ans (9)**

$$\begin{aligned}\text{Energy incident} &= \frac{hc}{\lambda} \\ &= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{248 \times 10^{-9} \times 1.6 \times 10^{-19}} \text{ eV} \\ &= \frac{6.63 \times 3 \times 100}{248 \times 1.6} \\ &= 0.05 \text{ eV} \times 100 = 5 \text{ eV}\end{aligned}$$

Now using

$$E = \phi + \text{K.E.}$$

$$5 = 3 + \text{K.E.}$$

$$\text{K.E.} = 2 \text{ eV} = 3.2 \times 10^{-19} \text{ J}$$

$$\text{for de-Broglie wavelength } \lambda = \frac{h}{mv}$$

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$\text{so } v = \sqrt{\frac{2\text{KE}}{m}}$$

$$\begin{aligned}\text{hence } \lambda &= \frac{h}{\sqrt{2\text{KE} \times m}} \\ &= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 3.2 \times 10^{-19} \times 9.1 \times 10^{-31}}} \\ &= \frac{6.63}{7.6} \times \frac{10^{-34}}{10^{-25}} = \frac{66.3 \times 10^{-10} \text{ m}}{7.6} \\ &= 8.72 \times 10^{-10} \text{ m} \\ &\approx 9 \times 10^{-10} \text{ m} = 9 \text{ \AA}\end{aligned}$$

8. **Ans (50)**

For A_2X



$$K_{sp} = 4S_1^3 = 4 \times 10^{-12}$$

$$S_1 = 10^{-4}$$

for MX



$$K_{sp} = S_2^2 = 4 \times 10^{-12}$$

$$S_2 = 2 \times 10^{-6}$$

$$\text{so } \frac{S_{A_2X}}{S_{MX}} = \frac{10^{-4}}{2 \times 10^{-6}} = 50$$

9. **Ans (33)**

$$\text{For BCC } \sqrt{3} a = 4r$$

$$\text{so } r = \frac{\sqrt{3}}{4} \times 27$$

for FCC

$$a = 2\sqrt{2} r$$

$$= 2 \times \sqrt{2} \times \frac{\sqrt{3}}{4} \times 27$$

$$= \frac{\sqrt{3}}{\sqrt{2}} \times 27$$

$$= 33$$

10. **Ans (10)**

$$K_{300} = 10^{-4} \quad K_{200} = ?$$

$$E_a = 11.488 \text{ KJ/mole} \quad R = 8.314 \text{ J/mole-K}$$

$$\text{so } \ln \left(\frac{K_{300}}{K_{200}} \right) = \frac{E_a}{R} \left(\frac{1}{200} - \frac{1}{300} \right)$$

$$\ln \left(\frac{K_{300}}{K_{200}} \right) = \frac{11.488 \times 1000 \times 100}{8.314 \times 200 \times 300}$$

$$= 2.303 = \ln 10$$

$$\text{so } \frac{K_{300}}{K_{200}} = 10$$

$$K_{200} = \frac{1}{10} \times K_{300} = 10^{-4}$$

$$= 10 \times 10^{-5} \text{ sec}^{-1}$$

PART-3 : MATHEMATICS

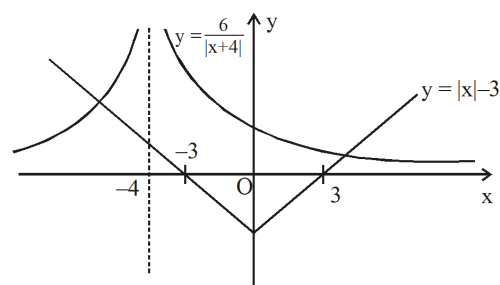
SECTION-I

1. **Ans (B)**

$$x \neq -4$$

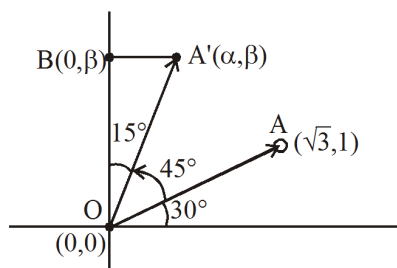
$$(|x| - 3)(|x + 4|) = 6$$

$$\Rightarrow |x| - 3 = \frac{6}{|x + 4|}$$



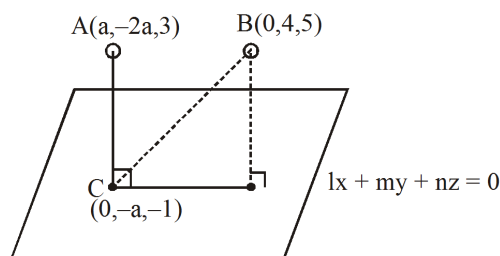
No. of solutions = 2

2. Ans (A)



$$\begin{aligned}\text{Area of } \triangle(OA'B) &= \frac{1}{2} OA' \cos 15^\circ \times OA' \sin 15^\circ \\ &= \frac{1}{2} (OA')^2 \frac{\sin 30^\circ}{2} \\ &= (3+1) \times \frac{1}{8} = \frac{1}{2}\end{aligned}$$

3. Ans (D)



$$\begin{aligned}C \text{ lies on plane} &\Rightarrow -ma - n = 0 \Rightarrow \frac{m}{n} = -\frac{1}{a} \\ \dots(1)\end{aligned}$$

$$\vec{CA} \parallel \ell \hat{i} + m \hat{j} + n \hat{k}$$

$$\frac{a-0}{\ell} = \frac{-a}{m} = \frac{4}{n} \Rightarrow \frac{m}{n} = -\frac{a}{4} \dots(2)$$

From (1) & (2)

$$-\frac{1}{a} = \frac{-a}{4} \Rightarrow a^2 = 4 \Rightarrow a = 2 \text{ (since } a > 0)$$

$$\text{From (2)} \quad \frac{m}{n} = \frac{-1}{2}$$

$$\text{Let } m = -t \Rightarrow n = 2t$$

$$\frac{2}{\ell} = \frac{-2}{-t} \Rightarrow \ell = t$$

$$\text{So plane : } t(x - y + 2z) = 0$$

$$BD = \frac{6}{\sqrt{6}} = \sqrt{6} \quad C \cong (0, -2, -1)$$

$$\begin{aligned}CD &= \sqrt{BC^2 - BD^2} \\ &= \sqrt{(0^2 + 6^2 + 6^2) - (\sqrt{6})^2} \\ &= \sqrt{66}\end{aligned}$$

4. Ans (D)

For a, b, c

$$\text{mean} = \frac{a+b+c}{3} (= \bar{x})$$

$$b = a + c$$

$$\Rightarrow \bar{x} = \frac{2b}{3} \dots(1)$$

$$\text{S.D. } (a+2, b+2, c+2) = \text{S.D. } (a, b, c) = d$$

$$\Rightarrow d^2 = \frac{a^2 + b^2 + c^2}{3} - (\bar{x})^2$$

$$\Rightarrow d^2 = \frac{a^2 + b^2 + c^2}{3} - \frac{4b^2}{9}$$

$$\Rightarrow 9d^2 = 3(a^2 + b^2 + c^2) - 4b^2$$

$$\Rightarrow b^2 = 3(a^2 + c^2) - 9d^2$$

5. Ans (B)

$$x \in \left(0, \frac{\pi}{2}\right)$$

$$\log_{10} \sin x + \log_{10} \cos x = -1$$

$$\Rightarrow \log_{10} \sin x \cdot \cos x = -1$$

$$\Rightarrow \sin x \cdot \cos x = \frac{1}{10} \dots(1)$$

$$\log_{10} (\sin x + \cos x) = \frac{1}{2} (\log_{10} n - 1)$$

$$\Rightarrow \sin x + \cos x = 10^{\left(\log_{10} \sqrt{n} - \frac{1}{2}\right)} = \sqrt{\frac{n}{10}}$$

by squaring

$$1 + 2 \sin x \cdot \cos x = \frac{n}{10}$$

$$\Rightarrow 1 + \frac{1}{5} = \frac{n}{10} \Rightarrow n = 12$$

6. Ans (C)

$$A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = 2 \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A^4 = 2^2 \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 8 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^8 = 64 \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = 128 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$$

$$\Rightarrow 128 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$$

$$\Rightarrow 128 \begin{bmatrix} x-y \\ -x+y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$$

$$\Rightarrow x - y = \frac{1}{16} \dots(1)$$

$$\& -x + y = \frac{1}{2} \dots(2)$$

$\Rightarrow$ From (1) & (2) : No solution.

7. **Ans (D)**

For standard parabola

For more than 3 normals (on axis)

$$x > \frac{L}{2} \text{ (where L is length of L.R.)}$$

$$\text{For } y^2 = 2x$$

$$\text{L.R.} = 2$$

for (a, 0)

$$a > \frac{\text{L.R.}}{2} \Rightarrow a > 1$$

8. **Ans (B)**

$$P(3, -1, 2)$$

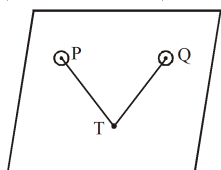
$$Q(1, 2, -4)$$

$$\overrightarrow{PR} \parallel 4\hat{i} - \hat{j} + 2\hat{k}$$

$$\overrightarrow{QS} \parallel -2\hat{i} + \hat{j} - 2\hat{k}$$

dr's of normal to the plane containing P, T & Q will be proportional to :

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 2 \\ -2 & 1 & -2 \end{vmatrix}$$



$$\therefore \frac{\ell}{0} = \frac{m}{4} = \frac{n}{2}$$

$$\text{For point, T : } \overrightarrow{PT} = \frac{x-3}{4} = \frac{y+1}{-1} = \frac{z-2}{2} = \lambda$$

$$\overrightarrow{QT} = \frac{x-1}{-2} = \frac{y-1}{1} = \frac{z+4}{-2} = \mu$$

$$T : (4\lambda + 3, -\lambda - 1, 2\lambda + 2) \\ \cong (2\mu + 1, \mu + 2, -2\mu - 4)$$

$$4\lambda + 3 = -2\mu + 1 \Rightarrow 2\lambda + \mu = -1$$

$$\lambda + \mu = -3 \Rightarrow \lambda = 2$$

$$\& \mu = -5 \quad \lambda + \mu = -3 \Rightarrow \lambda = 2$$

So point T : (11, -3, 6)

$$\overrightarrow{OA} = (11\hat{i} - 3\hat{j} + 6\hat{k}) \pm \left(\frac{2\hat{j} + \hat{k}}{\sqrt{5}} \right) \sqrt{5}$$

$$\overrightarrow{OA} = (11\hat{i} - 3\hat{j} + 6\hat{k}) \pm (2\hat{j} + \hat{k})$$

$$\overrightarrow{OA} = 11\hat{i} - \hat{j} + 7\hat{k}$$

or

$$9\hat{i} - 5\hat{j} + 5\hat{k}$$

$$|\overrightarrow{OA}| = \sqrt{121 + 1 + 49} = \sqrt{171}$$

or

$$\sqrt{81 + 25 + 25} = \sqrt{131}$$

9. **Ans (B)**

$$f(g(x)) = \begin{cases} g(x) + 2, & g(x) < 0 \\ (g(x))^2, & g(x) \geq 0 \end{cases}$$

$$= \begin{cases} x^3 + 2, & x < 0 \\ x^6, & x \in [0, 1) \\ (3x - 2)^2, & x \in [1, \infty) \end{cases}$$

$$(f \circ g(x))' = \begin{cases} 3x^2, & x < 0 \\ 6x^5, & x \in (0, 1) \\ 2(3x - 2) \times 3, & x \in (1, \infty) \end{cases}$$

At 'O'

L.H.L. $\neq$ R.H.L. (Discontinuous)

At '1'

$$\text{L.H.D.} = 6 = \text{R.H.D.}$$

$\Rightarrow f \circ g(x)$ is differentiable for $x \in \mathbb{R} - \{0\}$

10. **Ans (D)**

p	q	$p \wedge q$	$p \rightarrow q$	$(p \wedge q) \rightarrow (p \rightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

$(p \wedge q) \rightarrow (p \rightarrow q)$ is tautology

11. **Ans (B)**

$$\log_{\frac{1}{\sqrt{2}}} \left(\frac{|z| + 11}{(|z| - 1)^2} \right) \leq 2$$

$$\frac{|z| + 11}{(|z| - 1)^2} \geq \frac{1}{2}$$

$$2|z| + 22 \geq (|z| - 1)^2$$

$$2|z| + 22 \geq |z|^2 + 1 - 2|z|$$

$$|z|^2 - 4|z| - 21 \leq 0$$

$$\Rightarrow |z| \leq 7$$

$\therefore$ Largest value of $|z|$ is 7

12. **Ans (A)**
 $(3^{1/4} + 5^{1/8})^{60}$
 ${}^{60}C_r (3^{1/4})^{60-r} \cdot (5^{1/8})^r$
 ${}^{60}C_r (3)^{\frac{60-r}{4}} \cdot 5^{\frac{r}{8}}$

For rational terms.

$$\frac{r}{8} = k ; 0 \leq r \leq 60$$

$$0 \leq 8k \leq 60$$

$$0 \leq k \leq \frac{60}{8}$$

$$0 \leq k \leq 7.5$$

$$k = 0, 1, 2, 3, 4, 5, 6, 7$$

$$\frac{60-8k}{4} \text{ is always divisible by 4 for all value of } k.$$

Total rational terms = 8

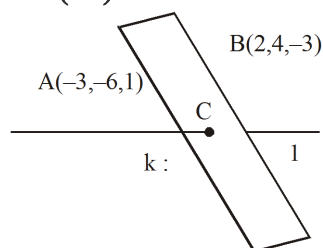
Total terms = 61

irrational terms = 53

$$n-1 = 53-1 = 52$$

52 is divisible by 26.

13. **Ans (C)**



Point C is

$$\left(\frac{2k-3}{k+1}, \frac{4k-6}{k+1}, \frac{-3k+1}{k+1} \right)$$

$$\frac{x-1}{-1} = \frac{y+4}{2} = \frac{z+2}{3}$$

Plane $\ell x + my + nz = 0$

$$\ell(-1) + m(2) + n(3) = 0$$

$$-\ell + 2m + 3n = 0 \quad \dots(1)$$

It also satisfy point $(1, -4, -2)$

$$\ell - 4m - 2n = 0 \quad \dots(2)$$

Solving (1) and (2)

$$2m + 3n = 4m + 2n$$

$$n = 2m$$

$$\ell - 4m - 4m = 0$$

$$\ell = 8m$$

$$\frac{\ell}{8} = \frac{m}{1} = \frac{n}{2}$$

$$\ell : m : n = 8 : 1 : 2$$

Plane is $8x + y + 2z = 0$

It will satisfy point C

$$8 \left(\frac{2k-3}{k+1} \right) + \left(\frac{4k-6}{k+1} \right) + 2 \left(\frac{-3k+1}{k+1} \right) = 0$$

$$16k - 24 + 4k - 6 - 6k + 2 = 0$$

$$14k = 28 \quad \therefore k = 2$$

14. **Ans (B)**

$$f(x) = (4a-3)(x + \log_e 5) + (a-7)\sin x$$

$$f(x) = (4a-3)(1) + (a-7)\cos x = 0$$

$$\Rightarrow \cos x = \frac{3-4a}{a-7}$$

$$-1 \leq \frac{3-4a}{a-7} < 1$$

$$\frac{3-4a}{a-7} + 1 \geq 0 \quad \frac{3-4a}{a-7} < 1$$

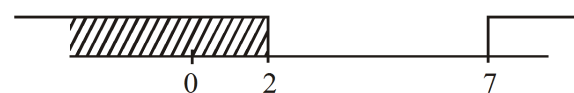
$$\frac{3-4a+a-7}{a-7} \geq 0 \quad \frac{3-4a}{a-7} - 1 < 0$$

$$\frac{-3a-4}{a-7} \geq 0 \quad \frac{3-4a-a+7}{a-7} < 0$$

$$\frac{3a+4}{a-7} \leq 0 \quad \frac{-5a+10}{a-7} < 0$$

$$\frac{5a-10}{a-7} > 0$$

$$\frac{5(a-2)}{a-7} > 0$$



$$\alpha \in \left[-\frac{4}{3}, 2 \right)$$

$$\text{Check end point } \left[-\frac{4}{3}, 2 \right)$$

15. **Ans (C)**

E_1 : Event denotes spade is missing

$$P(E_1) = \frac{1}{4}; P(\bar{E}_1) = \frac{3}{4}$$

A : Event drawn two cards are spade

$$P(A) = \frac{\frac{1}{4} \times \left(\frac{{}^{12}C_2}{{}^{51}C_2} \right) + \frac{3}{4} \times \left(\frac{{}^{13}C_2}{{}^{51}C_2} \right) + \frac{3}{4} \times \left(\frac{{}^{13}C_2}{{}^{51}C_2} \right)}{\frac{1}{4} \times \left(\frac{{}^{12}C_2}{{}^{51}C_2} \right) + \frac{3}{4} \times \left(\frac{{}^{13}C_2}{{}^{51}C_2} \right)}$$

$$= \frac{39}{50}$$

16. Ans (C)

$$(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$$

$$(1 - x + x^3)^n = a_0 + a_1 x + a_2 x^2 \dots + a_{3n} x^{3n}$$

$$\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} = \text{Sum of } a_0 + a_2 + a_4 \dots$$

$$\sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1} = \text{Sum of } a_1 + a_3 + a_5 \dots$$

put $x = 1$

$$1 = a_0 + a_1 + a_2 + a_3 \dots + a_{3n} \dots (A)$$

Put $x = -1$

$$1 = a_0 - a_1 + a_2 - a_3 \dots + (-1)^{3n} a_{3n} \dots (B)$$

Solving (A) and (B)

$$a_0 + a_2 + a_4 \dots = 1$$

$$a_1 + a_3 + a_5 \dots = 0$$

$$\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1} = 1$$

17. Ans (D)

$$\frac{dy}{dx} + 2y \tan x = \sin x$$

$$\text{I.F.} = e^{\int 2 \tan x \, dx} = e^{2 \ln \sec x}$$

$$\text{I.F.} = \sec^2 x$$

$$y \cdot (\sec^2 x) = \int \sin x \cdot \sec^2 x \, dx$$

$$y \cdot (\sec^2 x) = \int \sec x \tan x \, dx$$

$$y \cdot (\sec^2 x) = \sec x + C$$

$$x = \frac{\pi}{3}; y = 0$$

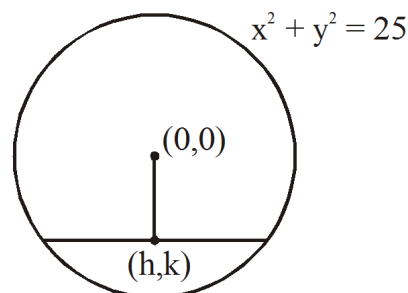
$$\Rightarrow C = -2$$

$$\Rightarrow y = \frac{\sec x - 2}{\sec^2 x} = \cos x - 2 \cos^2 x$$

$$y = t - 2t^2 \Rightarrow \frac{dy}{dt} = 1 - 4t = 0 \Rightarrow t = \frac{1}{4}$$

$$\therefore \max = \frac{1}{4} - \frac{1}{8} = \frac{2-1}{8} = \frac{1}{8}$$

18. Ans (D)



Equation of chord

$$y - k = -\frac{h}{k}(x - h)$$

$$ky - k^2 = -hx + h^2$$

$$hx + ky = h^2 + k^2$$

$$y = -\frac{hx}{k} + \frac{h^2 + k^2}{k}$$

$$\text{tangent to } \frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$c^2 = a^2 m^2 - b^2$$

$$\left(\frac{h^2 + k^2}{k}\right)^2 = 9\left(-\frac{h}{k}\right)^2 - 16$$

$$(x^2 + y^2)^2 = 9x^2 - 16y^2$$

19. Ans (B)

$$(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$$

$$(81)^{\sin^2 x} + \frac{(81)^1}{(18)^{\sin^2 x}} = 30$$

$$(81)^{\sin^2 x} = t$$

$$t + \frac{81}{t} = 30$$

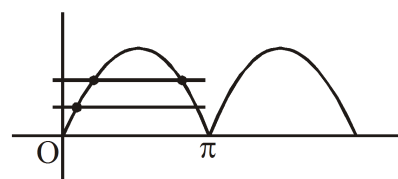
$$t^2 - 30t + 81 = 0$$

$$(t - 3)(t - 27) = 0$$

$$(81)^{\sin^2 x} = 3^1 \quad \text{or} \quad (81)^{\sin^2 x} = 3^3$$

$$3^{4 \sin^2 x} = 3^1 \quad \text{or} \quad 3^{4 \sin^2 x} = 3^3$$

$$\sin^2 x = \frac{1}{4} \quad \text{or} \quad \sin^2 x = \frac{3}{4}$$



Total sol. = 4

PART-3 : MATHEMATICS

SECTION-II

20. Ans (C)

$$S_k = \sum_{r=1}^k \tan^{-1} \left(\frac{6^r}{2^{2r+1} + 3^{2r+1}} \right)$$

Divide by 3^{2r}

$$\sum_{r=1}^k \tan^{-1} \left(\frac{\left(\frac{2}{3}\right)^r}{\left(\frac{2}{3}\right)^{2r} + 3} \right)$$

$$\sum_{r=1}^k \tan^{-1} \left(\frac{\left(\frac{2}{3}\right)^r}{3 \left(\left(\frac{2}{3}\right)^{2r+1} + 1 \right)} \right)$$

$$\text{Let } \left(\frac{2}{3}\right)^r = t$$

$$\sum_{r=1}^k \tan^{-1} \left(\frac{\frac{t}{3}}{1 + \frac{2}{3}t^2} \right)$$

$$\sum_{r=1}^k \tan^{-1} \left(\frac{t - \frac{2t}{3}}{1 + t \cdot \frac{2t}{3}} \right)$$

$$\sum_{r=1}^k \left(\tan^{-1}(t) - \tan^{-1} \left(\frac{2t}{3} \right) \right)$$

$$\sum_{r=1}^k \left(\tan^{-1} \left(\frac{2}{3} \right)^r - \tan^{-1} \left(\frac{2}{3} \right)^{r+1} \right)$$

$$S_k = \tan^{-1} \left(\frac{2}{3} \right) - \tan^{-1} \left(\frac{2}{3} \right)^{k+1}$$

$$S_{\infty} = \lim_{k \rightarrow \infty} \left(\tan^{-1} \left(\frac{2}{3} \right) - \tan^{-1} \left(\frac{2}{3} \right)^{k+1} \right)$$

$$= \tan^{-1} \left(\frac{2}{3} \right) - \tan^{-1} (0)$$

$$\therefore S_{\infty} = \tan^{-1} \left(\frac{2}{3} \right) = \cot^{-1} \left(\frac{3}{2} \right)$$

1. Ans (3)

GP : 4, 8, 16, 32, 64, 128, 256, 512, 1024,

2048, 4096, 8192

AP : 11, 16, 21, 26, 31, 36

Common terms : 16, 256, 4096 only

2. Ans (1)

$$E = 2 \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} f \left(\frac{r}{n} \right)$$

$$E = \frac{2}{\ln 2} \int_0^1 \ln \left(1 + \tan \frac{\pi x}{4} \right) dx \quad \dots(i)$$

replacing $x \rightarrow 1 - x$

$$E = \frac{2}{\ln 2} \int_0^1 \ln \left(1 + \tan \frac{\pi}{4} (1 - x) \right) dx$$

$$E = \frac{2}{\ln 2} \int_0^1 \ln \left(1 + \tan \left(\frac{\pi}{4} - \frac{\pi}{4} x \right) \right) dx$$

$$E = \frac{2}{\ln 2} \int_0^1 \ln \left(1 + \frac{1 + \tan \frac{\pi}{4} x}{1 + \tan \frac{\pi}{4} x} \right) dx$$

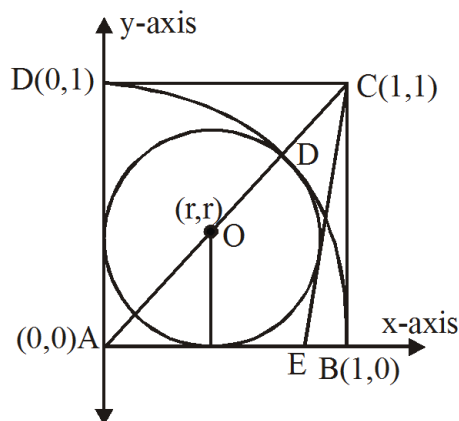
$$E = \frac{2}{\ln 2} \int_0^1 \ln \left(\frac{2}{1 + \tan \frac{\pi x}{4}} \right) dx$$

$$E = \frac{2}{\ln 2} \int_0^1 \left(\ln 2 - \ln \left(1 + \tan \frac{\pi x}{4} \right) \right) dx \quad \dots(ii)$$

equation (i) + (ii)

$$E = 1$$

3. Ans (1)



Here $AO + OD = 1$ or $(\sqrt{2} + 1)r = 1$

$$\Rightarrow r = \sqrt{2} - 1$$

$$\text{equation of circle } (x - r)^2 + (y - r)^2 = r^2$$

Equation of CE

$$y - 1 = m(x - 1)$$

$$mx - y + 1 - m = 0$$

It is tangent to circle

$$\therefore \left| \frac{mr - r + 1 - m}{\sqrt{m^2 + 1}} \right| = r$$

$$\left| \frac{(m - 1)r + 1 - m}{\sqrt{m^2 + 1}} \right| = r$$

$$\frac{(m - 1)^2 (r - 1)^2}{m^2 + 1} = r^2$$

$$\text{Put } r = \sqrt{2} - 1$$

$$\text{On solving } m = 2 - \sqrt{3}, 2 + \sqrt{3}$$

Taking greater slope of CE as

$$2 + \sqrt{3}$$

$$y - 1 = (2 + \sqrt{3})(x - 1)$$

$$\text{Put } y = 0$$

$$-1 = (2 + \sqrt{3})(x - 1)$$

$$\frac{-1}{2 + \sqrt{3}} \times \left(\frac{2 - \sqrt{3}}{2 - \sqrt{3}} \right) = x - 1$$

$$x - 1 = \sqrt{3} - 1$$

$$EB = 1 - x = 1 - (\sqrt{3} - 1)$$

$$EB = 2 - \sqrt{3}$$

4. Ans (4)

$$\lim_{x \rightarrow 0} \frac{ae^x - b \cos x + ce^{-x}}{x \sin x} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{a \left(1 + x + \frac{x^2}{2!} \dots \right) - b \left(1 - \frac{x^2}{2!} + \dots \right) + c \left(1 - x + \frac{x^2}{2!} \right)}{\left(\frac{x \sin x}{x} \right) x} = 2$$

$$a - b + c = 0 \quad \dots (1)$$

$$a - c = 0 \quad \dots (2)$$

$$\& \frac{a + b + c}{2} = 2$$

$$\Rightarrow \boxed{a + b + c = 4}$$

5. Ans (766)

$$\text{Let } A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

diagonal elements of

$$AA^T, a^2 + b^2 + c^2, d^2 + e^2 + f^2, g^2 + h^2 + i^2$$

$$\text{Sum} = a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 9$$

$$a, b, c, d, e, f, g, h, i \in \{0, 1, 2, 3\}$$

	Case	No. of Matrices
(1)	All - 1s	$\frac{9!}{9!} = 1$
(2)	One $\rightarrow$ 3 remaining-0	$\frac{9!}{1! \times 8!} = 9$
(3)	One-2 five- 1s three-0s	$\frac{9!}{1! \times 5! \times 3!} = 8 \times 63$
(4)	two - 2's one- 1 six-0's	$\frac{9!}{2! \times 6!} = 63 \times 4$

$$\text{Total no. of ways} = 1 + 9 + 8 \times 63 + 63 \times 4$$

$$= \boxed{766}$$

6. Ans (36)

$$\text{Let } M = (P^{-1}AP - I)^2$$

$$= (P^{-1}AP)^2 - 2P^{-1}AP + I$$

$$= P^{-1}A^2P - 2P^{-1}AP + I$$

$$PM = A^2P - 2AP + P$$

$$= (A^2 - 2A.I + I^2)P$$

$$\Rightarrow \text{Det}(PM) = \text{Det}((A - I)^2 \times P)$$

$$\Rightarrow \text{Det}P \cdot \text{Det}M = \text{Det}(A - I)^2 \times \text{Det}(P)$$

$$\Rightarrow \text{Det } M = (\text{Det}(A - I))^2$$

$$\text{Now } A - I = \begin{bmatrix} 1 & 7 & \omega^2 \\ -1 & -\omega - 1 & 1 \\ 0 & -\omega & -\omega \end{bmatrix}$$

$$\text{Det}(A - I) = (\omega^2 + \omega + \omega) + 7(-\omega) + \omega^3 = -6\omega$$

$$\text{Det}((A - I)^2) = 36\omega^2$$

$$\Rightarrow \alpha = 36$$

7. Ans (406)

$$y(x) = \int_0^x (2t^2 - 15t + 10) dt$$

$$y'(x)]_{x=a} = [2x^2 - 15x + 10]_a = 2a^2 - 15a + 10$$

$$\text{Slope of normal} = -\frac{1}{3}$$

$$\Rightarrow 2a^2 - 15a + 10 = 3 \Rightarrow a = 7$$

$$\& a = \frac{1}{2} \text{ (rejected)}$$

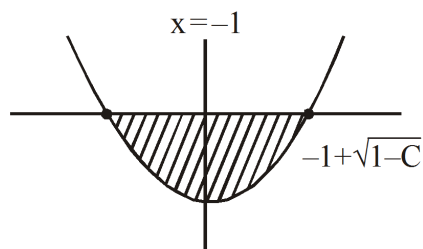
$$b = y(7) = \int_0^7 (2t^2 - 15t + 10) dt$$

$$= \left[\frac{2t^3}{3} - \frac{15t^2}{2} + 10t \right]_0^7$$

$$\Rightarrow 6b = 4 \times 7^3 - 45 \times 49 + 60 \times 7$$

$$|a + 6b| = 406$$

8. Ans (2)



$$\frac{dy}{dx} = 2(x+1)$$

$$\Rightarrow \int dy = \int 2(x+1) dx$$

$$\Rightarrow y(x) = x^2 + 2x + C$$

$$\text{Area} = \frac{4\sqrt{8}}{3}$$

$$-1 + \sqrt{1-C}$$

$$\Rightarrow 2 \int_{-1}^{-1+\sqrt{1-C}} (-(x+1)^2 - C + 1) dx = \frac{4\sqrt{8}}{3}$$

$$\Rightarrow 2 \left[-\frac{(x+1)^3}{3} - Cx + x \right]_{-1}^{-1+\sqrt{1-C}} = \frac{4\sqrt{8}}{3}$$

$$\Rightarrow -(\sqrt{1-C})^3 + 3C - 3C\sqrt{1-C} - 3 + 3\sqrt{1-C} - 3C + 3 = 2\sqrt{8}$$

$$\Rightarrow C = -1$$

$$\Rightarrow f(x) = x^2 + 2x - 1, f(1) = 2$$

9. Ans (16)

$$f(x) + f(x+1) = 2$$

$$\Rightarrow f(x) \text{ is periodic with period} = 2$$

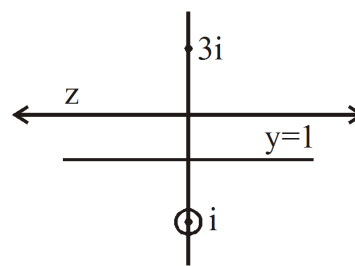
$$I_1 = \int_0^8 f(x) dx = 4 \int_0^2 f(x) dx$$

$$= 4 \int_0^1 (f(x) + f(1+x)) dx = 8$$

$$\text{Similarly } I_2 = 2 \times 2 = 4$$

$$I_1 + 2I_2 = 16$$

10. Ans (4)



$$w = z\bar{z} - 2z + 2$$

$$\left| \frac{z+i}{z-3i} \right| = 1$$

$$\Rightarrow |z+i| = |z-3i|$$

$$\Rightarrow z = x + i, x \in \mathbb{R}$$

$$w = (x+i)(x-i) - 2(x+i) + 2$$

$$= x^2 + 1 - 2x - 2i + 2$$

$$\text{Re}(w) = x^2 - 2x + 3$$

$$\text{For min (Re(w)), } x = 1$$

$$\Rightarrow w = 2 - 2i = 2(1-i) = 2\sqrt{2} e^{-i\frac{\pi}{4}}$$

$$w^n = (2\sqrt{2})^n e^{-i\frac{n\pi}{4}}$$

$$\text{For real \& minimum value of } n,$$

$$n = 4$$